

MODULE – III

Virtual circuits and datagrams. Principles of routing. Internet protocol Ipv4 CIDR. Routing algorithms: Link-state and distance vector routing. Routing on the internet RIP, OSPF and BGP, Multicast routing. Introduction to IPV6

Introduction to Network Layer

The Network Layer is the third layer of the OSI (Open Systems Interconnection) model and is responsible for delivering data packets from the source device to the destination device across multiple networks. It determines the best path for data transmission and ensures that packets travel efficiently through intermediate routers.

Unlike the Data Link Layer, which operates within a single network, the Network Layer enables communication between different networks, making it fundamental for the functioning of the Internet.

The most widely used protocol at this layer is the Internet Protocol (IP), which assigns logical addresses to devices and facilitates packet forwarding.

Functions of the Network Layer

Logical Addressing

Logical addressing is used to uniquely identify devices in a network.

- The Network Layer assigns **IP addresses** to each device.
- These addresses help routers determine where a packet should be sent.

Routing

Routing is the process of **selecting the best path** for data packets to travel from the source to the destination. Routers maintain **routing tables** that store information about network paths.

Common routing approaches include:

- **Static Routing** – routes are manually configured.
- **Dynamic Routing** – routers automatically update routes using algorithms.

Important routing protocols include:

- **Routing Information Protocol (RIP)**
- **Open Shortest Path First (OSPF)**
- **Border Gateway Protocol (BGP)**

Packet Forwarding

Forwarding refers to the **actual movement of packets from one router to another**.

Steps involved:

1. Router receives the packet.
2. The router examines the destination IP address.
3. The routing table is consulted.
4. Packet is forwarded to the next hop router.

This process continues until the packet reaches the destination network.

Congestion Control

The Network Layer also deals with **network congestion**, which occurs when too many packets compete for limited bandwidth.

Methods used include:

- Traffic shaping
- Packet dropping
- Congestion avoidance algorithms

Effective congestion management improves overall network performance.

Services Provided by the Network Layer

The Network Layer provides two main types of services:

Connectionless Service

- Data packets are sent **independently**.
- Each packet may take a **different route**.
- No prior connection is established.

Example: Internet communication using Internet Protocol.

Connection-Oriented Service

- A **logical path is established before transmission**.
- All packets follow the same route.
- Reliability is improved.

Example: Virtual circuit networks.

Virtual Circuits and Datagrams

In computer networking, the Network Layer provides two major packet delivery approaches: **Virtual Circuit networks** and **Datagram networks**. These approaches define how packets travel from a source to a destination across interconnected networks. The difference lies mainly in **connection establishment, routing behaviour, reliability, and packet handling**.

1. Virtual Circuit Network

A Virtual Circuit (VC) is a connection-oriented communication method in which a logical path is established between the sender and receiver before any data transmission begins. Once the path is established, all packets follow the **same route** through the network until the communication session ends.

This mechanism is similar to a **telephone call**, where a dedicated path is created before communication starts.

Virtual Circuit Working Stages

The working of a virtual circuit occurs in **three major phases**:

1. Connection Establishment (Setup Phase)

Before data transmission begins, the sender must establish a virtual connection with the destination.

Step-by-step process

1. The source device sends a **connection request packet** to the network.
2. Routers along the path analyse the request.
3. Each router reserves resources and assigns a **Virtual Circuit Identifier (VCI)**.
4. The routers create entries in their **VC tables**.
5. Once the request reaches the destination, a **connection acknowledgment** is returned.

After this process, a **logical path is created from source to destination**.

2. Data Transfer Phase

After the connection is established, the sender starts transmitting data packets.

Working process:

1. The sender divides the message into packets.
2. Each packet carries the **Virtual Circuit Identifier (VCI)** instead of a full address.
3. Routers read the VCI value.
4. Using the VC table, the router forwards the packet to the next router.
5. All packets follow the **same predetermined route**.

Important features:

- No routing decision is required during transmission.
- Packets reach the destination **in correct order**.
- Less processing overhead at routers.

3. Connection Termination (Release Phase)

After data transmission is complete:

1. The sender sends a **connection termination request**.
2. Routers delete the VC entries from their tables.
3. Allocated resources are released.

This completes the communication session.

Characteristics of Virtual Circuits

1. Connection-oriented communication.
2. Same path used for all packets.
3. Requires setup phase before data transfer.
4. Packets arrive in correct order.
5. Routers maintain **state information**.

2. Datagram Network

A **Datagram Network** is a **connectionless communication system** where **each packet is treated independently**. There is **no predetermined path**, and each packet may travel through **different routes** to reach the destination.

The Internet primarily uses this approach through Internet Protocol.

Operation of Datagram Networks

Steps in packet delivery:

1. Source device sends packets independently.
2. Each packet contains the **complete destination address**.
3. Routers examine the destination address.
4. The best route is chosen dynamically.
5. Packets may arrive **out of order**.

Working Process of Datagram Networks

In datagram communication, the network does not maintain a dedicated connection between devices.

1. **Packet Creation**
 - The source divides the message into packets.
 - Each packet contains the **complete destination IP address**.
2. **Independent Packet Transmission**
 - Packets are transmitted individually.
 - Each packet may travel through different routes.
3. **Routing Decision at Each Router**
 - When a packet arrives at a router, the router checks the **destination IP address**.
 - The router consults its **routing table**.
 - It selects the best available route.

4. Forwarding to Next Router

- The packet is forwarded to the next hop router.
- This process continues until the packet reaches the destination.

5. Packet Reassembly

- Packets may arrive **out of order**.
- The destination system reassembles them using sequence numbers.

Characteristics of Datagram Networks

1. Connectionless communication.
2. No setup phase required.
3. Each packet routed independently.
4. Packets may take different paths.
5. Routers do **not maintain connection state**.

Feature	Virtual Circuit	Datagram
Connection Type	Connection-oriented	Connectionless
Path	Fixed path	Dynamic path
Setup Phase	Required	Not required
Packet Order	Guaranteed	Not guaranteed
Router State	Maintained	Not maintained
Addressing	VC Identifier	Full destination address
Example	ATM, Frame Relay	Internet (IP)

Routing

- A Routing is a process of **selecting path** along which the data can be transferred from source to the destination. Routing is performed by a special device known as a **router**.
- A Router works at the **network layer in the OSI model and internet layer in TCP/IP model**
- A router is a networking device that forwards the packet based on the information available in the packet header and forwarding table.
- The **routing algorithms** are used for routing the packets. The routing algorithm is nothing but a software responsible for deciding the optimal path through which packet can be transmitted.
- The **routing protocols** use the metric to determine the best path for the packet delivery. The metric is the standard of measurement such as hop count, bandwidth, delay, current load on the path, etc. used by the routing algorithm to determine the optimal path to the destination.
- The routing algorithm initializes and maintains the routing table for the process of path determination.

Routing Metrics and Costs

- Routing metrics and costs are used for determining the best route to the destination. The factors used by the protocols to determine the shortest path; these factors are known as a **metric**.
- Metrics are the **network variables** used to determine the best route to the destination.

Most common metric values:

- **Hop count:** Hop count is a fundamental routing metric used in computer networks to measure the number of intermediate devices (routers) through which a data packet must pass to reach its destination. Each time a packet moves from one router to another, it is considered one **hop**. Therefore, hop count represents the total number of routing steps between the source and destination.

A hop is counted only when the packet passes through a router.

The source computer and destination computer are not counted as hops.

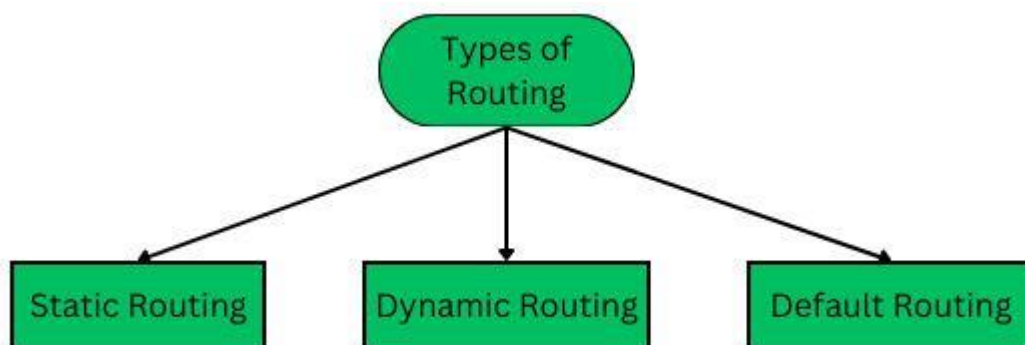
More routers in the path means higher hop count.

Source Computer → Router 1 → Router 2 → Router 3 → Destination Computer

Hop count = 3

- **Delay:** The total time taken for a packet to move from the source node to the destination node across a communication network.
- **Bandwidth:** The maximum amount of data that can be transmitted over a communication channel in a specified time interval.
- **Load:** **Load** is an important routing metric used in computer networks to represent the amount of traffic currently present on a network link or router.
- **Reliability:** In routing decisions, reliability indicates how consistently a communication link can transmit data without failures, packet loss, or transmission errors.

Types of Routing



Static Routing

Static routing is a routing method in which the network administrator manually enters the route information into the router. The router does not learn routes automatically. Once the route is configured, the router will always use that path unless the administrator changes it. Static routing means fixed routing paths that are manually controlled.

How Static Routing Works

1. The network administrator configures the route manually in the router.
2. The router stores the information in its routing table.
3. When a data packet arrives, the router checks the destination IP address.
4. The router compares the destination with entries in the routing table.
5. If a matching route exists, the packet is forwarded to the next-hop router or exit interface.

For example, suppose a router needs to send data to another network. The administrator specifies that all packets for that network must be sent through a specific router. The router will always follow this path.

Characteristics of Static Routing

- Routes are configured manually by an administrator.
- The router does not exchange routing information with other routers.
- Routing paths remain fixed and stable.
- Suitable for simple network structures.

Advantages of Static Routing

1. Simplicity – Configuration is straightforward for small networks.
2. Low resource usage – No routing protocol traffic is generated.
3. Better security – Routes are not automatically shared with other routers.
4. Predictable network behaviour – The routing path remains constant.

Disadvantages of Static Routing

1. Lack of flexibility – The router cannot automatically adjust if the network changes.
2. Manual maintenance – The administrator must update routes whenever the topology changes.
3. Limited scalability – Not practical for large networks with many routers.

Dynamic Routing

Dynamic routing is a routing technique where routers automatically discover, exchange, and update routing information with other routers. This is done through routing protocols.

Instead of manually entering routes, routers communicate with each other and determine the best path automatically.

Examples of routing protocols include:

- Routing Information Protocol
- Open Shortest Path First
- Border Gateway Protocol

How Dynamic Routing Works

1. Routers run a routing protocol.
2. Routers share information about networks they know with neighbouring routers.
3. Each router builds its own routing table based on this information.
4. Routing algorithms calculate the best path using metrics such as:
 - Hop count (number of routers)
 - Delay
 - Bandwidth
 - Reliability
5. If a network link fails or a new route appears, routers automatically update their routing tables.

For example, if one network path becomes unavailable, the router will automatically select another path without administrator intervention.

Characteristics of Dynamic Routing

- Routes are learned automatically.
- Routers exchange routing information periodically.
- Uses routing algorithms to find the best path.
- Suitable for large and complex networks.

Advantages of Dynamic Routing

1. Automatic route updates when network changes occur.
2. Scalability for large networks.
3. Reduced administrative effort.
4. Efficient path selection based on routing metrics.

Disadvantages of Dynamic Routing

1. Higher complexity in configuration and management.
2. Consumes network bandwidth due to routing updates.
3. Requires more processing power and memory in routers.

Default Routing

Default routing is a routing technique used when a router does not know the exact path to the destination network. In such cases, the router sends the packet to a default router (called the default gateway), which then forwards the packet toward the correct destination.

In simple words, default routing is a fall back path used when no specific route exists in the routing table.

Every router maintains a routing table that contains information about different network paths. When a packet arrives:

1. The router checks the routing table for the destination network.
2. If a matching route is found, the packet is forwarded through that route.
3. If no matching route exists, the router uses the default route.

This means the router should send packets for any unknown network to the default gateway.

How Default Routing Works

Consider a small network connected to the internet.

Computer → Router → Internet

1. A computer sends a packet to a website.
2. The router checks its routing table.
3. If the destination network is not listed in the routing table:
4. The router forwards the packet to the default gateway.
5. The gateway router then determines the correct path.

Thus, default routing allows routers to forward packets even when they do not know the exact destination route.

Characteristics of Default Routing

- Used when no specific route is available.
- Sends packets to a default gateway router.
- Simplifies routing table management.
- Often used in small networks connected to a larger network.

Advantages

1. Simplifies routing tables because many specific routes are not required.
2. Easy configuration for small networks.
3. Useful when a network has one main exit path, such as an internet connection.

Disadvantages

1. The path chosen may not always be the most efficient.
2. If the default gateway fails, communication may stop.
3. Overuse in large networks can cause inefficient routing decisions.

Internet Protocol

- An IP stands for **Internet Protocol**.
- An IP address is **assigned to each device** connected to a network. Each device uses an **IP address** for communication.
- It also behaves as an **identifier** as this address is used to identify the device on a network.

- Internet Protocol (IP) is a fundamental communication protocol used in computer networks to identify devices and deliver data packets from a source device to a destination device across interconnected networks.
- It operates at the network layer of the TCP/IP model and is responsible for logical addressing and routing of packets across the internet.
- The primary purpose of Internet Protocol is to ensure that **data sent from one device can reach another device on a network or across multiple networks.**
- To facilitate the **routing of packets**, TCP/IP protocol uses a 32-bit logical address known as IPv4 (Internet Protocol version 4).

An IP address consists of two parts:

1. **Network address**
2. **Host address.**

1. Network Part (Network ID)

The **network part** identifies the **specific network** in which the device is located. All devices within the same network share the **same network portion** of the IP address.

- It tells routers **which network the packet should be sent to.**
- Routers use this part to **route data between different networks.**
- Devices in the same network have the **same network ID but different host IDs.**

Example

Consider the IPv4 address:

192.168.1.25

If the subnet mask indicates that **192.168.1** is the network portion, then:

- **Network ID:** 192.168.1
- This identifies the **network segment.**

2. Host Part (Host ID)

The **host part** identifies the **specific device within the network.**

- Each device inside a network must have a **unique host ID.**
- It allows the network to distinguish between **different devices such as computers, printers, or servers.**

Example

Using the same IP address:

192.168.1.25

- **Host ID:** 25

- This identifies the **specific device inside the network**.

IP Address = Network ID + Host ID

Example:

192.168.1 | 25

- **192.168.1** → Network part
- **25** → Host part

Structure of an IP Packet

An IP packet consists of two main parts:

1. Header

The header contains control information such as:

- Source IP address
- Destination IP address
- Packet length
- Fragmentation information
- Time to live (TTL)

2. Data (Payload)

This part contains the **actual data being transmitted**, such as messages, files, or web content.

Versions of Internet Protocol

Internet Protocol (IP) is used to **identify devices and transmit data across networks**. The two main versions of IP are **IPv4** and **IPv6**. These protocols define how devices are addressed and how packets are routed in a network.

IPv4

IPv4 is a network layer protocol that uses a 32-bit logical addressing scheme to uniquely identify devices connected to a network. These addresses allow routers to deliver packets to the correct destination across multiple networks.

Structure of IPv4 Address

An IPv4 address contains **32 bits**, which are divided into **four groups called octets**. Each octet consists of **8 bits**.

The address is written in **decimal format separated by dots**.

Example:

192.168.1.10

Structure:

8 bits 8 bits 8 bits 8 bits
192 . 168 . 1 . 10

Each octet can have a value between:

0 – 255

Parts of IPv4 Address

An IPv4 address consists of **two logical components**.

1. Network ID

The **network portion** identifies the network to which the device belongs. Routers use this part to determine **which network the packet should be delivered to**.

2. Host ID

The **host portion** identifies the specific device within the network.

Example:

192.168.1.25

- Network ID → 192.168.1
- Host ID → 25

This structure allows millions of devices to communicate while maintaining a hierarchical addressing system.

- In today's computer network world, computers do not understand the IP addresses in the standard numeric format as the computers understand the numbers in binary form only.
- The binary number can be either 1 or 0. The IPv4 consists of four sets, and these sets represent the octet. The bits in each octet represent a number.
- Each bit in an octet can be either 1 or 0. If the bit the 1, then the number it represents will count, and if the bit is 0, then the number it represents does not count.

Representation of 8 Bit Octet

128	64	32	16	8	4	2	1
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Now, we will see how to obtain the binary representation of the above IP address, i.e., 66.94.29.13

Step 1: First, we find the binary number of 66.

128	64	32	16	8	4	2	1
0	1	0	0	0	0	1	0

To obtain 66, we put 1 under 64 and 2 as the sum of 64 and 2 is equal to 66 ($64+2=66$), and the remaining bits will be zero, as shown above. Therefore, the binary bit version of 66 is 01000010.

Step 2: Now, we calculate the binary number of 94.

128	64	32	16	8	4	2	1
0	1	0	1	1	1	1	0

To obtain 94, we put 1 under 64, 16, 8, 4, and 2 as the sum of these numbers is equal to 94, and the remaining bits will be zero. Therefore, the binary bit version of 94 is 01011110.

Step 3: The next number is 29.

128	64	32	16	8	4	2	1
0	0	0	1	1	1	0	0

To obtain 29, we put 1 under 16, 8, 4, and 1 as the sum of these numbers is equal to 29, and the remaining bits will be zero. Therefore, the binary bit version of 29 is 00011101.

Step 4: The last number is 13.

128	64	32	16	8	4	2	1
0	0	0	0	1	1	0	1

To obtain 13, we put 1 under 8, 4, and 1 as the sum of these numbers is equal to 13, and the remaining bits will be zero. Therefore, the binary bit version of 13 is 00001101

Features of IPv4

IPv4 provides several important features:

- **32-bit addressing system**
- Supports approximately **4.3 billion unique addresses**
- Uses **dotted decimal notation**
- Supports **packet fragmentation**
- Compatible with most existing networking systems

IPv4 PACKET STRUCTURE

Version	Header Length	Type Of Service	
Total Length		Flags	Fragment Offset
Identification		Header Checksum	
Source IP Address			
Destination IP Address			

- **Version (4 bits)** → IP version (IPv4 = 4)
- **Header Length (4 bits)** → Size of header
- **Type of Service (8 bits)** → Priority (speed, delay, reliability)
- **Total Length (16 bits)** → Total packet size (header + data)
- **Identification (16 bits)** → ID for packet fragments
- **Flags (3 bits)** → Control fragmentation
- **Fragment Offset (13 bits)** → Position of fragment
- **Time To Live (8 bits)** → Limits number of hops
- **Protocol (8 bits)** → Next protocol (TCP/UDP)
- **Header Checksum (16 bits)** → Error check for header
- **Source IP Address (32 bits)** → Sender address
- **Destination IP Address (32 bits)** → Receiver address
- **Options (variable)** → Extra info (optional)

Limitations of IPv4

Although IPv4 played a crucial role in the development of the internet, several limitations emerged over time:

1. **Limited address space** due to the rapid growth of internet users and devices.
2. Need for **Network Address Translation (NAT)** to conserve addresses.
3. **Complex configuration and management** in large networks.
4. Limited built-in security features.

IPv6

- IPv4 produces 4 billion addresses, and the developers think that these addresses are enough, but they were wrong.
- IPv6 is the next generation of IP addresses.

- The main difference between IPv4 and IPv6 is the address size of IP addresses.
- The IPv4 is a 32-bit address, whereas IPv6 is a 128-bit hexadecimal address.
- IPv6 provides a large address space, and it contains a simple header as compared to IPv4.
- 128-bit Address Architecture

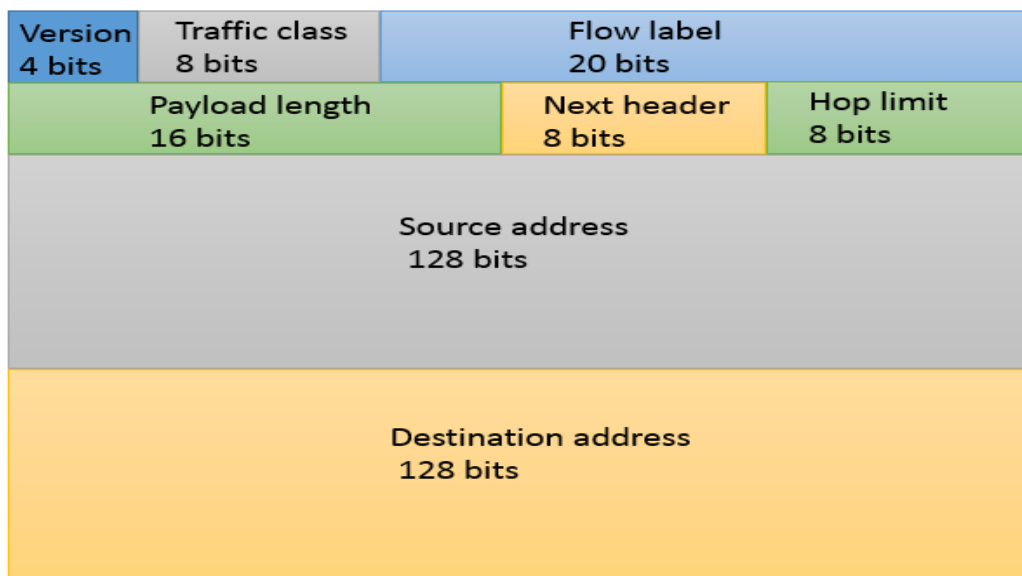
IPv6 uses a **128-bit binary address**, conceptually divided into:

- **Network Prefix (first 64 bits)** → identifies the network
- **Interface Identifier (last 64 bits)** → identifies the host/device

Representation in Hexadecimal

IPv6 addresses are expressed in hexadecimal form:

2001:0db8:0000:0000:0000:ff00:0042:8329



- **Version (4-bits):** Indicates version of Internet Protocol which contains bit sequence 0110.
- **Traffic Class (8-bits):** The Traffic Class field indicates class or priority of IPv6 packet which is similar to *Service Field* in IPv4 packet. It helps routers to handle the traffic based on the priority of the packet. If congestion occurs on the router then packets with the least priority will be discarded.

As of now, only 4-bits are being used (and the remaining bits are under research), in which 0 to 7 are assigned to Congestion controlled traffic and 8 to 15 are assigned to Uncontrolled traffic.

Priority assignment of Congestion controlled traffic:

Priority	Meaning
0	No Specific traffic
1	Background data
2	Unattended data traffic
3	Reserved
4	Attended bulk data traffic
5	Reserved
6	Interactive traffic
7	Control traffic

- Uncontrolled data traffic is mainly used for Audio/Video data. So we give higher priority to uncontrolled data traffic.
- The source node is allowed to set the priorities but on the way, routers can change it. Therefore, the destination should not expect the same priority which was set by the source node.
- **Flow Label (20-bits):** Flow Label field is used by a source to label the packets belonging to the same flow in order to request special handling by intermediate IPv6 routers, such as non-default quality of service or real-time service. In order to distinguish the flow, an intermediate router can use the source address, a destination address, and flow label of the packets. Between a source and destination, multiple flows may exist because many processes might be running at the same time. Routers or Host that does not support the functionality of flow label field and for default router handling, flow label field is set to 0. While setting up the flow label, the source is also supposed to specify the lifetime of the flow.
- **Payload Length (16-bits):** It is a 16-bit (unsigned integer) field, indicates the total size of the payload which tells routers about the amount of information a particular packet contains in its payload.
- **Next Header (8-bits):** Next Header indicates the type of extension header (if present) immediately following the IPv6 header. Whereas In some cases it indicates the protocols contained within upper-layer packets, such as TCP, UDP.
- **Hop Limit (8-bits):** Hop Limit field is the same as TTL in IPv4 packets. It indicates the maximum number of intermediate nodes IPv6 packet is allowed to travel. Its value gets decremented by one, by each node that forwards the packet and the packet is discarded if the value decrements to 0. This is used to discard the packets that are stuck in an infinite loop because of some routing error.

- **Source Address (128-bits):** Source Address is the 128-bit IPv6 address of the original source of the packet.
- **Destination Address (128-bits):** The destination Address field indicates the IPv6 address of the final destination (in most cases). All the intermediate nodes can use this information in order to correctly route the packet.

Address format for IPv6



IPv6 is a 128-bit hexadecimal address. It contains 8 fields separated by a colon, and each field is 16-bit in size.

Difference between IPv4 and IPv6

	IPv4	IPv6
Address length	IPv4 is a 32-bit address.	IPv6 is a 128-bit address.
Fields	IPv4 is a numeric address that consists of 4 fields which are separated by dot (.).	IPv6 is an alphanumeric address that consists of 8 fields, which are separated by colon.
Classes	IPv4 has 5 different classes of IP address that includes Class A, Class B, Class C, Class D, and Class E.	IPv6 does not contain classes of IP addresses.
Number of IP address	IPv4 has a limited number of IP addresses.	IPv6 has a large number of IP addresses.
VLSM	It supports VLSM (Virtual Length Subnet Mask). Here, VLSM means that Ipv4 converts IP addresses into a subnet of different sizes.	It does not support VLSM.
Address configuration	It supports manual and DHCP configuration.	It supports manual, DHCP, auto-configuration, and renumbering.
Address space	It generates 4 billion unique addresses.	It generates 340 undecillion unique addresses.

End-to-end connection integrity	In IPv4, end-to-end connection integrity is unachievable.	In the case of IPv6, end-to-end connection integrity is achievable.
Security features	In IPv4, security depends on the application. This IP address is not developed in keeping the security feature in mind.	In IPv6, IPSEC is developed for security purposes.
Address representation	In IPv4, the IP address is represented in decimal.	In IPv6, the representation of the IP address in hexadecimal.
Fragmentation	Fragmentation is done by the senders and the forwarding routers.	Fragmentation is done by the senders only.
Packet flow identification	It does not provide any mechanism for packet flow identification	It uses flow label field in the header for the packet flow identification.
Checksum field	The checksum field is available in IPv4.	The checksum field is not available in IPv6.
Transmission scheme	IPv4 is broadcasting.	On the other hand, IPv6 is multicasting, which provides efficient network operations.
Encryption and Authentication	It does not provide encryption and authentication.	It provides encryption and authentication.
Number of octets	It consists of 4 octets.	It consists of 8 fields, and each field contains 2 octets. Therefore, the total number of octets in IPv6 is 16.

CLASSFUL ADDRESSING

- The IP address comprises up of 32 bits and is split into four sections separated by dots: **part 1, part 2, part 3, and part 4.**
- The IP address is made up of four parts, each of which is eight bits long (1 byte).
- Further, the 4 parts of the IP address is divided into parts: a **network ID** and a **Host ID**.
- In classful addressing, the IP address space is divided into five classes: A, B, C, D, and E.
- **Each class occupies some part of the whole address space.**

Finding the class of address

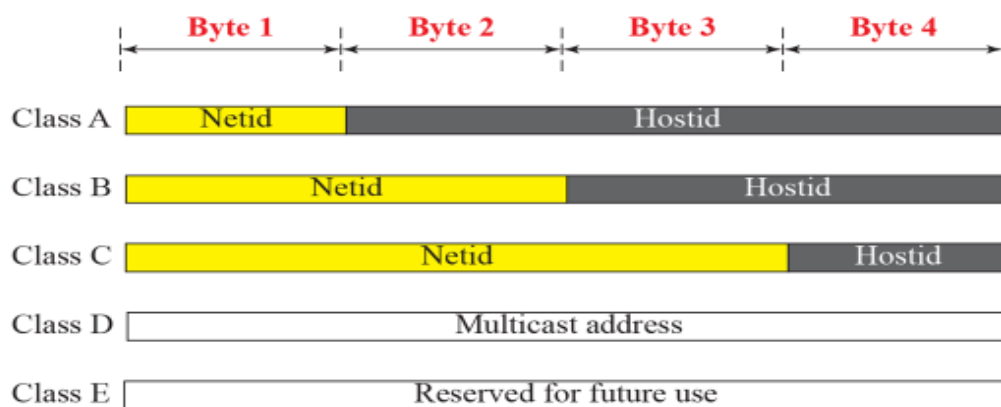
We can find the class of an address when the address is given either in binary or dotted decimal notation. In the binary notation, the first few bits can immediately tell us the class of the address; in the dotted-decimal notation, the value of the first byte can give the class of an address.

	Octet 1	Octet 2	Octet 3	Octet 4
Class A	0.....			
Class B	10.....			
Class C	110.....			
Class D	1110....			
Class E	1111....			
Binary notation				
	Byte 1	Byte 2	Byte 3	Byte 4
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–299			
Class E	240–255			
Dotted-decimal notation				

Problems with Classful Addressing:

- The problem with this classful addressing method is that millions of class A address are wasted, many of the class B address are wasted, whereas, number of addresses available in class C is so small that it cannot cater the needs of organizations. Class D addresses are used for multicast routing and are therefore available as a single block only. Class E addresses are reserved.

Since there are these problems, Classful networking was replaced by Classless Inter-Domain Routing (CIDR) in 1993



CLASSLESS INTER-DOMAIN ROUTING

CIDR is a method used in computer networks to assign IP addresses in a flexible and efficient way. It removes the concept of fixed classes (like Class A, B, C) and allows networks to be designed based on actual requirements. CIDR helps use IP addresses properly without wasting them.

CIDR Notation

CIDR uses a format:

IP address / prefix length

- The **prefix length** tells how many bits are used for the network
- The remaining bits are used for devices (hosts)

Working of CIDR

Division of Address

CIDR divides an IP address into:

- Network part
- Host part

This division is **not fixed** and can be changed depending on the need.

Flexible Allocation

CIDR allocates IP addresses based on the number of devices required.

It gives a block of addresses that closely matches the requirement, avoiding unnecessary allocation.

Routing Mechanism

CIDR uses a concept called **longest prefix matching**:

- Routers compare destination addresses with available network prefixes

- The **most specific match** (largest prefix) is selected

This improves routing accuracy.

Route Aggregation

CIDR allows combining multiple networks into a single route.

- This reduces the number of entries in routing tables
- Improves overall network performance

Features of CIDR

Flexibility

- No fixed classes
- Network size can be adjusted as needed

Efficient Address Usage

- Reduces wastage of IP addresses
- Allocates only what is required

Smaller Routing Tables

- Supports route aggregation
- Reduces memory and processing load on routers

Scalability

- Suitable for large and growing networks
- Supports expansion of the Internet

Improved Performance

- Faster routing decisions
- Better network management

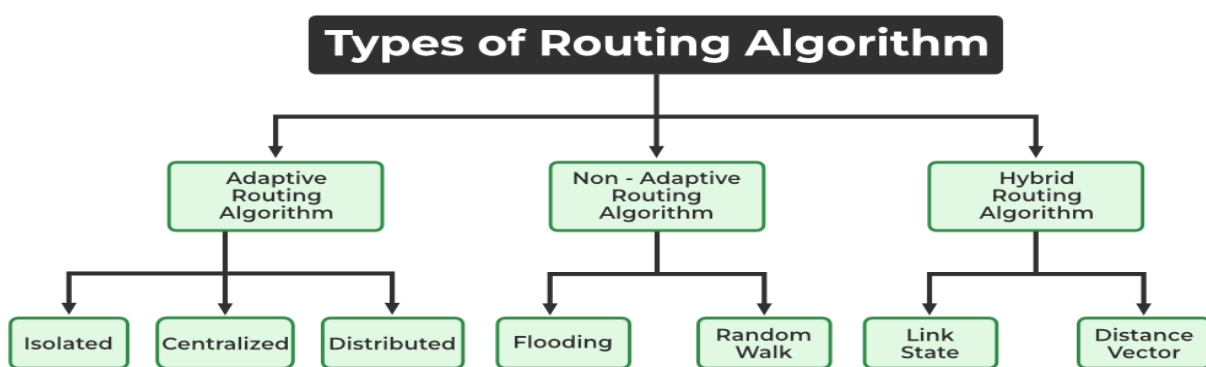
CIDR is a modern and efficient IP addressing method that replaces classful addressing. It provides flexibility, reduces wastage, and improves routing efficiency.

PARAMETERS	CLASSFULL	CIDR
Basics	IP addresses are divided into fixed classes (A, B, C, D, E)	IP addresses are allocated without fixed classes using variable-length prefixes

Practical Use	Less practical due to rigid address allocation	More practical and flexible in real networks
Network ID & Host ID	Network and host portions are fixed based on the class	Network and host portions are defined by the prefix length
Subnetting	Subnetting is not supported by default	Subnetting is inherently supported
VLSM	Does not support Variable Length Subnet Mask (VLSM)	Supports VLSM
IP Address Utilization	Poor utilization; leads to significant IP address wastage	Efficient utilization of IP addresses
Routing Efficiency	Large routing tables; no route aggregation	Supports route aggregation, resulting in smaller routing tables
CIDR Support	Does not support CIDR	Fully supports CIDR
Scalability	Limited scalability due to fixed classes	Highly scalable for large and modern networks
Troubleshooting	Simpler but inefficient due to rigid structure	More efficient and manageable despite slightly higher complexity
Division of Address	Network ID and Host ID only	Network prefix and Host ID (flexible boundary)

Routing Algorithms

- Routing algorithms are used in Computer Networking to determine the best path for data transmission.
 - They help data packets move from a source device to a destination efficiently.
 - The main purpose is to select the shortest or fastest route available.
 - They reduce network congestion by avoiding busy paths.
 - Routing algorithms ensure reliable delivery even if some links fail.
 - They continuously update routes based on network changes.
 - These algorithms use metrics like hop count, delay, and bandwidth.
 - They improve overall network performance and speed.
 - Routing algorithms also help in optimal use of network resources.
- Thus, they play a crucial role in smooth and effective communication between devices.



Distance vector routing algorithm

Distance Vector Routing is a fundamental dynamic routing approach in Computer Networking, where each router determines the best path based on **distance (cost) and direction (next hop)**. It is called “distance vector” because routers maintain a vector (table) of distances to all possible destinations.

In distance vector routing, each router shares its routing information only with its **immediate neighbors**, not the entire network. Based on this exchanged information, routers update their routing tables and gradually learn the best paths.

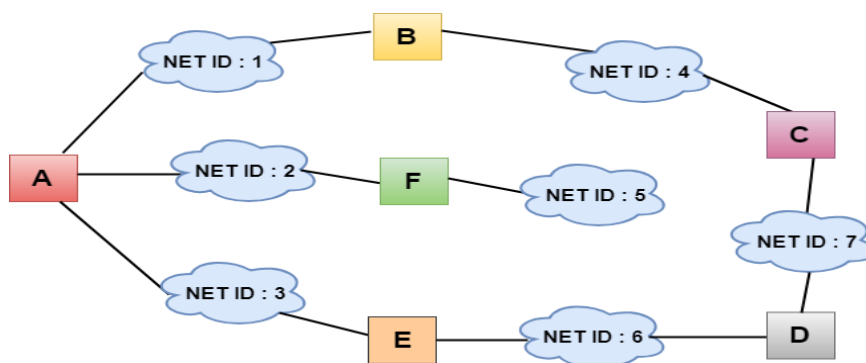
- Each router initially knows only directly connected networks.
- It creates a routing table containing:
 - Destination network
 - Distance (metric, e.g., hop count)
 - Next hop
- Routers periodically send their routing tables to neighboring routers.
- When a router receives an update, it compares the new route with its existing routes.
- It updates its table if a **shorter path** is found.
- This process continues until all routers have consistent routing information (convergence).

The Distance vector algorithm is iterative, asynchronous and distributed

- **Distributed:** It is distributed in that each node receives information from one or more of its directly attached neighbors, performs calculation and then distributes the result back to its neighbors.
- **Iterative:** It is iterative in that its process continues until no more information is available to be exchanged between neighbors.
- **Asynchronous:** It does not require that all of its nodes operate in the lock step with each other.
- The Distance vector algorithm is a **dynamic algorithm**.
- It is mainly used in **ARPANET**, and **RIP**.
- Each router maintains a distance table known as **Vector**.

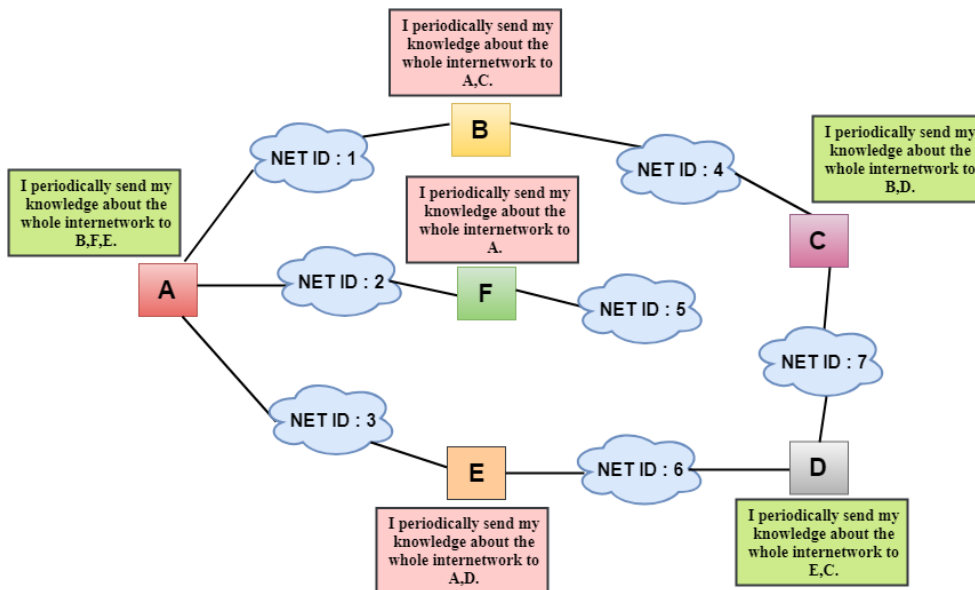
Three Keys to understand the working of Distance Vector Routing Algorithm:

- **Knowledge about the whole network:** Each router shares its knowledge through the entire network. The Router sends its collected knowledge about the network to its neighbors.
- **Routing only to neighbors:** The router sends its knowledge about the network to only those routers which have direct links. The router sends whatever it has about the network through the ports. The information is received by the router and uses the information to update its own routing table.
- **Information sharing at regular intervals:** Within 30 seconds, the router sends the information to the neighbouring routers.



- In the above figure, each cloud represents the network, and the number inside the cloud represents the network ID.
- All the LANs are connected by routers, and they are represented in boxes labelled as A, B, C, D, E, F.

- Distance vector routing algorithm simplifies the routing process by assuming the cost of every link is one unit. Therefore, the efficiency of transmission can be measured by the number of links to reach the destination.
- In Distance vector routing, the cost is based on hop count.



In the above figure, we observe that the router sends the knowledge to the immediate neighbors. The neighbors add this knowledge to their own knowledge and send the updated table to their own neighbors. In this way, routers get its own information plus the new information about the neighbors.

ROUTING TABLE

Two processes occur:

- Creating the Table
- Updating the Table

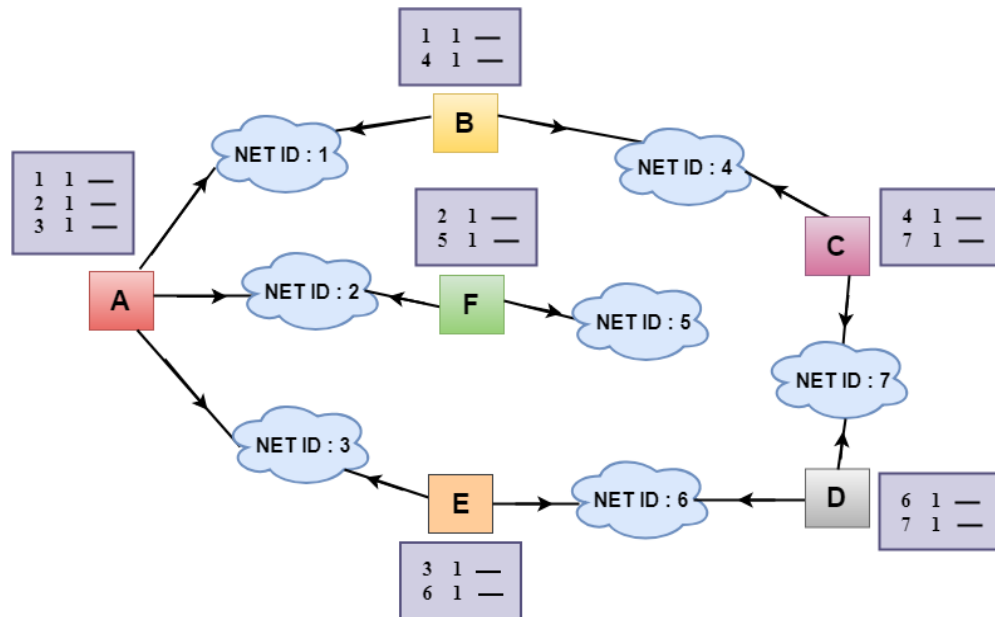
Creating the Table

Initially, the routing table is created for each router that contains at least three types of information such as Network ID, the cost and the next hop.

NET ID	Cost	Next Hop
---	---	---
---	---	---
---	---	---
---	---	---

- **NET ID:** The Network ID defines the final destination of the packet.
- **Cost:** The cost is the number of hops that packet must take to get there.

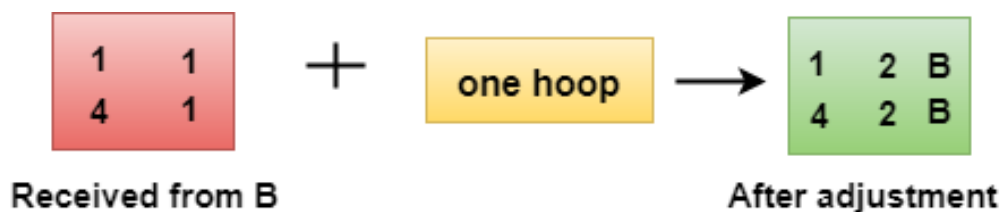
- **Next hop:** It is the router to which the packet must be delivered.



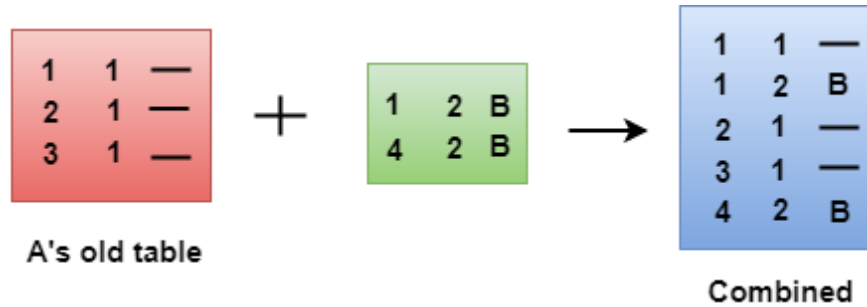
- In the above figure, the original routing tables are shown of all the routers. In a routing table, the first column represents the network ID, the second column represents the cost of the link, and the third column is empty.
- These routing tables are sent to all the neighbors.
- **For Example:**
 1. A sends its routing table to B, F & E.
 2. B sends its routing table to A & C.
 3. C sends its routing table to B & D.
 4. D sends its routing table to E & C.
 5. E sends its routing table to A & D.
 6. F sends its routing table to A.

Updating the Table

- When A receives a routing table from B, then it uses its information to update the table.
- The routing table of B shows how the packets can move to the networks 1 and 4.
- The B is a neighbor to the A router, the packets from A to B can reach in one hop. So, 1 is added to all the costs given in the B's table and the sum will be the cost to reach a particular network.

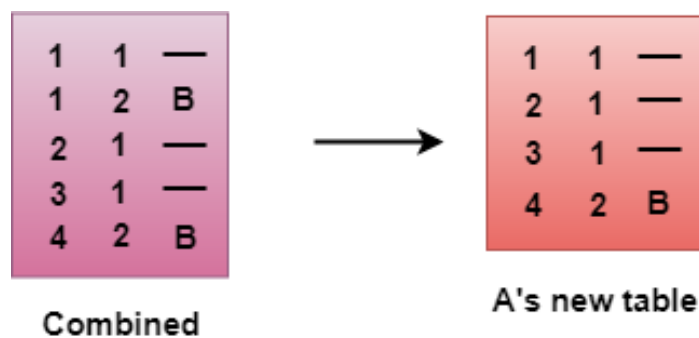


- After adjustment, A then combines this table with its own table to create a combined table.



- The combined table may contain some duplicate data. In the above figure, the combined table of router A contains the duplicate data, so it keeps only those data which has the lowest cost.

For example, A can send the data to network 1 in two ways. The first which uses no next router, so, it costs one hop. The second requires two hops (A to B, then B to Network 1). The first option has the lowest cost; therefore, it is kept and the second one is dropped.



- The process of creating the routing table continues for all routers. Every router receives the information from the neighbors, and updates the routing table.

Final routing tables of all the routers are given below:

Router A	Router B	Router C
6 2 E	6 3 E	6 2 D
1 1 -	1 1 -	1 2 B
3 1 -	3 2 A	3 3 D
4 2 B	4 1 -	4 1 -
7 3 E	7 2 C	7 1 -
2 1 -	2 2 A	2 3 B
5 2 F	5 3 A	5 4 B

Router D	Router E	Router F
6 1 -	6 1 -	6 3 A
1 3 E	1 2 A	1 2 A
3 2 E	3 1 -	3 2 A
4 2 C	4 3 A	4 3 A
7 1 -	7 2 D	7 4 A
2 3 E	2 2 A	2 1 -
5 4 E	5 3 A	5 1 -

LINK STATE ROUTING

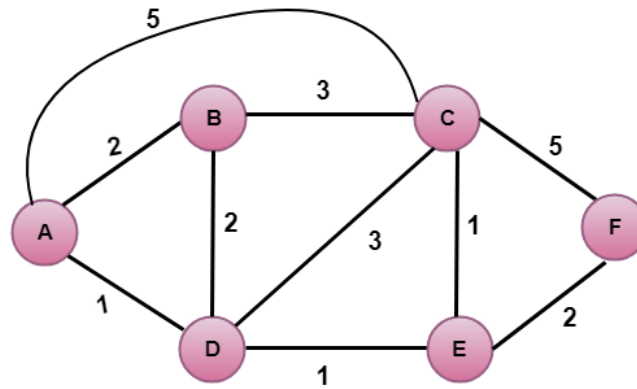
Link state routing is a technique in which each router shares the knowledge of its neighbourhood with every other router in the internetwork.

The three keys to understand the Link State Routing algorithm:

- **Knowledge about the neighbourhood:** Instead of sending its routing table, a router sends the information about its neighbourhood only. A router broadcast its identities and cost of the directly attached links to other routers.
- **Flooding:** Each router sends the information to every other router on the internetwork except its neighbors. This process is known as Flooding. Every router that receives the packet sends the copies to all its neighbors. Finally, each and every router receives a copy of the same information.
- **Information sharing:** A router sends the information to every other router only when the change occurs in the information.

Link-state routing is a dynamic routing technique used in computer networks where each router builds a complete map (topology) of the network and independently computes the shortest path to all other nodes. Unlike distance vector routing, which relies on neighbours' information, link-state routing enables routers to have a global view of the network.

Let's understand through an example:



In the above figure, source vertex is A.

Step 1: The first step is an initialization step. The currently known least cost path from A to its directly attached neighbors, B, C, D are 2,5,1 respectively. The cost from A to B is set to 2, from A to D is set to 1 and from A to C is set to 5. The cost from A to E and F are set to infinity as they are not directly linked to A.

Step	N	D(B), P(B)	D(C), P(C)	D(D), P(D)	D(E), P(E)	D(F), P(F)
1	A	2, A	5, A	1, A	∞	∞

Step 2: In the above table, we observe that vertex D contains the least cost path in step 1. Therefore, it is added in N. Now, we need to determine a least-cost path through D vertex.

a) Calculating shortest path from A to B

1. $v = B, w = D$
2. $D(B) = \min(D(B), D(D) + c(D,B))$
3. $= \min(2, 1+2)$
4. $= \min(2, 3)$
5. The minimum value is 2. Therefore, the currently shortest path from A to B is 2.

b) Calculating shortest path from A to C

1. $v = C, w = D$
2. $D(C) = \min(D(C), D(D) + c(D,C))$
3. $= \min(5, 1+3)$
4. $= \min(5, 4)$
5. The minimum value is 4. Therefore, the currently shortest path from A to C is 4

c) Calculating shortest path from A to E

1. $v = E, w = D$
2. $D(E) = \min(D(E), D(D) + c(D,E))$
3. $= \min(\infty, 1+1)$
4. $= \min(\infty, 2)$
5. The minimum value is 2. Therefore, the currently shortest path from A to E is 2.

{Note: The vertex D has no direct link to vertex E. Therefore, the value of $D(F)$ is infinity.}

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞

Step 3: In the above table, we observe that both E and B have the least cost path in step 2. Let's consider the E vertex. Now, we determine the least cost path of remaining vertices through E.

a) Calculating the shortest path from A to B.

1. $v = B, w = E$
2. $D(B) = \min(D(B), D(E) + c(E,B))$
3. $= \min(2, 2 + \infty)$
4. $= \min(2, \infty)$
5. The minimum value is 2. Therefore, the currently shortest path from A to B is 2.

b) Calculating the shortest path from A to C.

1. $v = C, w = E$
2. $D(C) = \min(D(C), D(E) + c(E,C))$
3. $= \min(4, 2+1)$
4. $= \min(4, 3)$
5. The minimum value is 3. Therefore, the currently shortest path from A to C is 3.

c) Calculating the shortest path from A to F.

1. $v = F, w = E$
2. $D(F) = \min(D(F), D(E) + c(E,F))$
3. $= \min(\infty, 2+2)$

$$4. = \min(\infty, 4)$$

5. The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E

Step 4: In the above table, we observe that B vertex has the least cost path in step 3. Therefore, it is added in N. Now, we determine the least cost path of remaining vertices through B.

a) Calculating the shortest path from A to C.

$$1. v = C, w = B$$

$$2. D(C) = \min(D(C), D(B) + c(B,C))$$

$$3. = \min(3, 2+3)$$

$$4. = \min(3, 5)$$

5. The minimum value is 3. Therefore, the currently shortest path from A to C is 3.

b) Calculating the shortest path from A to F.

$$1. v = F, w = B$$

$$2. D(F) = \min(D(F), D(B) + c(B,F))$$

$$3. = \min(4, \infty)$$

4. The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E

Step 5: In the above table, we observe that C vertex has the least cost path in step 4. Therefore, it is added in N. Now, we determine the least cost path of remaining vertices through C.

a) Calculating the shortest path from A to F.

1. $v = F, w = C$
2. $D(F) = \min(D(F), D(C) + c(C,F))$
3. $= \min(4, 3+5)$
4. $= \min(4,8)$
5. The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E
5	ADEBC					4,E

Final table:

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E
5	ADEBC					4,E
6	ADEBCF					

Shortest Distance from A

Node	Distance	Path
B	2	$A \rightarrow B$
C	3	$A \rightarrow D \rightarrow E \rightarrow C$
D	1	$A \rightarrow D$
E	2	$A \rightarrow D \rightarrow E$
F	4	$A \rightarrow D \rightarrow E \rightarrow F$

Shortest Path Tree

Edges:

- $A \rightarrow D$
- $A \rightarrow B$

- $D \rightarrow E$
- $E \rightarrow C$
- $E \rightarrow F$

Final Order of Nodes

$A \rightarrow D \rightarrow E \rightarrow B \rightarrow C \rightarrow F$

AUTONOMOUS SYSTEMS (ASS):

- An autonomous system (AS) is a large network or group of networks that has a unified routing policy. Every computer or device that connects to the Internet is connected to an AS.
- Routers within the same AS all run the same routing algorithm (For example, a LS or DV algorithm) and have information about each.
- The routing algorithm running within an autonomous system is called an **intra-autonomous system routing protocol**.
- It will be necessary, of course, to connect ASs to each other, and thus one or more of the routers in an AS will have the added task of being responsible for forwarding packets to destinations outside the AS;
- These routers are called **gateway routers**.

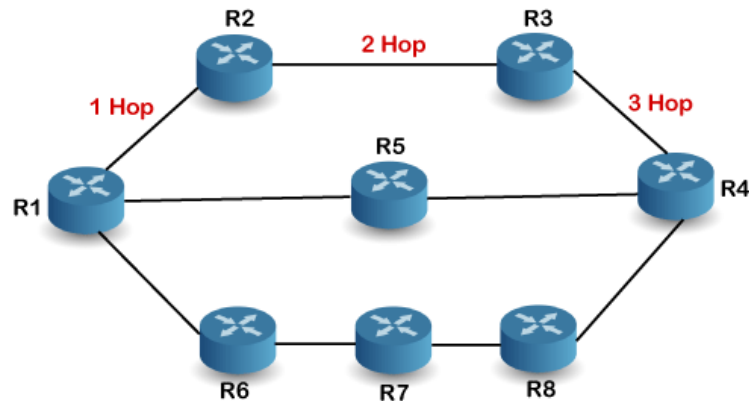
ROUTING ON THE INTERNET: RIP OSPF AND BGP

1. RIP PROTOCOL

- RIP stands for **Routing Information Protocol**.
- RIP is an **intra-domain routing protocol** used within an autonomous system. Here, intra-domain means routing the packets in a defined domain, for example, web browsing within an institutional area.
- RIP is based on the **distance vector-based strategy**, so we consider the entire structure as a graph where **nodes are the routers**, and the **links are the networks**.
- In a routing table, the first column is the destination, or we can say that it is a network address.
- The cost metric is the number of hops to reach the destination. The number of hops available in a network would be the cost. The hop count is the number of networks required to reach the destination.
- In RIP, infinity is defined as 16, which means that the RIP is useful for smaller networks or small autonomous systems. The maximum number of hops that RIP can contain is 15 hops, i.e., it should not have more than 15 hops as 16 is infinity.
- The next column contains the address of the router to which the packet is to be sent to reach the destination.

How is hop count determined?

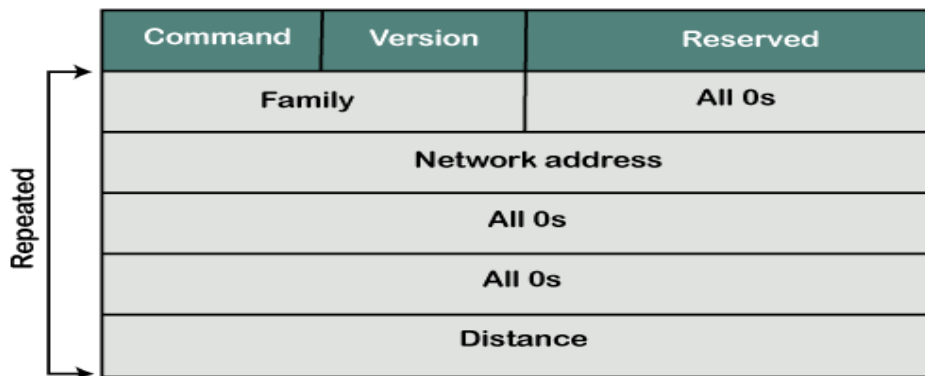
When the router sends the packet to the network segment, then it is counted as a single hop.



In the above figure, when the router 1 forwards the packet to the router 2 then it will count as 1 hop count. Similarly, when the router 2 forwards the packet to the router 3 then it will count as 2 hop count, and when the router 3 forwards the packet to router 4, it will count as 3 hop count. In the same way, [RIP](#) can support maximum up to 15 hops, which means that the 16 routers can be configured in a RIP.

RIP Message Format:

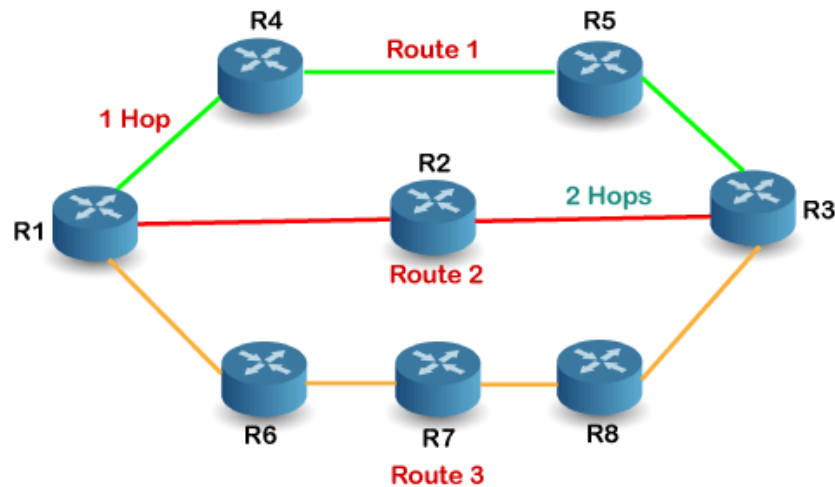
The message format is used to share information among different routers.



- **Command:** It is an 8-bit field that is used for request or reply. The value of the request is 1, and the value of the reply is 2.
- **Version:** Here, version means that which version of the protocol we are using. Suppose we are using the protocol of version1, then we put the 1 in this field.
- **Reserved:** This is a reserved field, so it is filled with zeroes.
- **Family:** It is a 16-bit field. As we are using the TCP/IP family, so we put 2 values in this field.
- **Network Address:** It is defined as 14 bytes field. If we use the IPv4 version, then we use 4 bytes, and the other 10 bytes are all zeroes.

- **Distance:** The distance field specifies the hop count, i.e., the number of hops used to reach the destination.

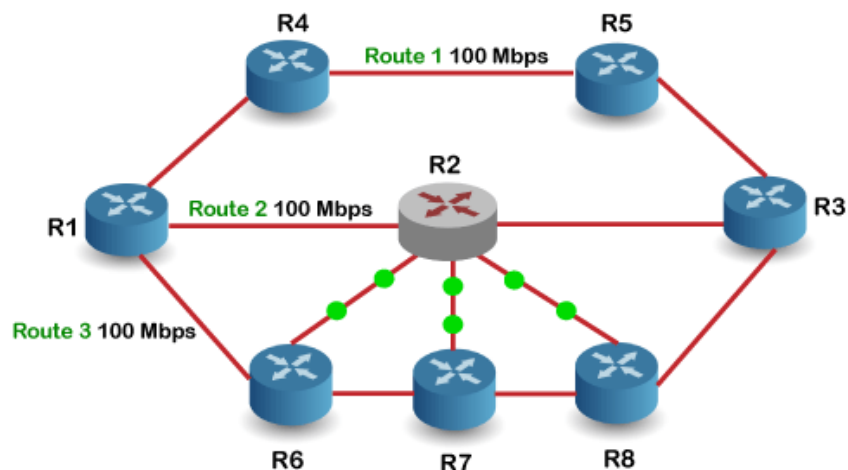
How does the RIP work?



If there are 8 routers in a network where, Router 1 wants to send the data to Router 3. If the network is configured with RIP, it will choose the route which has the least number of hops. There are three routes in the above network, i.e., Route 1, Route 2, and Route 3. The Route 2 contains the least number of hops, i.e., 2 where Route 1 contains 3 hops, and Route 3 contains 4 hops, so RIP will choose Route 2.

Disadvantages of RIP:

- In RIP, the route is chosen based on the hop count metric. If another route of better bandwidth is available, then that route would not be chosen.



We can observe that Route 2 is chosen in the above figure as it has the least hop count. The Route 1 is free and data can be reached faster; instead of this, data is sent to the Route 2 that makes the Route 2 slower due to the heavy traffic. This is one of the biggest disadvantages of RIP.

- It faces a problem of slow convergence. Whenever the router or link fails, then it often takes minutes to stabilize or take an alternative route; this problem is known as Slow convergence.
- RIP supports maximum 15 hops which means that the maximum 16 hops can be configured in a RIP
- The Administrative distance value is 120 (Ad value). If the Ad value is less, then the protocol is more reliable than the protocol with more Ad value. The RIP protocol has the highest Ad value, so it is not as reliable as the other routing protocols.

How RIP updates its Routing table

- **RIP update timer: 30 sec**

The routers configured with RIP send their updates to all the neighboring routers every 30 seconds.

- **RIP Invalid timer: 180 sec**

The RIP invalid timer is 180 seconds, which means that if the router is disconnected from the network or some link goes down, then the neighbor router will wait for 180 seconds to take the update. If it does not receive the update within 180 seconds, then it will mark the particular route as not reachable.

- **RIP Flush timer: 240 sec**

The RIP flush timer is 240 second which is almost equal to 4 min means that if the router does not receive the update within 240 seconds then the neighbor route will remove that particular route from the routing table which is a very slow process as 4 minutes is a long time to wait.

Advantages of RIP

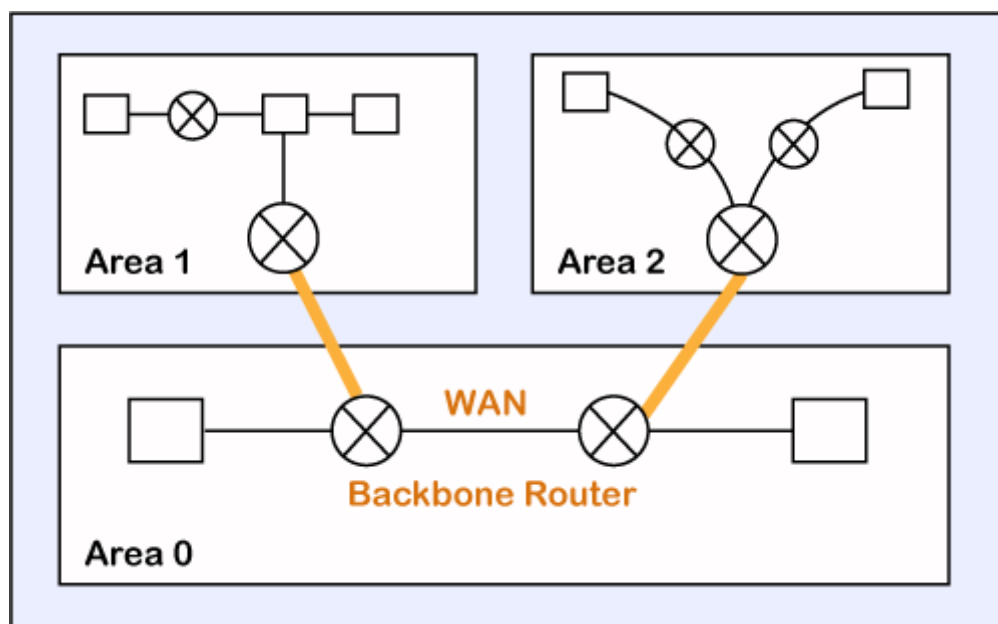
- It is easy to configure.
- It has less complexity
- The CPU utilization is less.

2. OSPF PROTOCOL

- The OSPF stands for **Open Shortest Path First**.
- It is a widely used and supported routing protocol.
- It is an ***intradomain protocol***, which means that it is used within an area or a network.
- It is an ***interior gateway protocol*** that has been designed within a single autonomous system.

- It is based on a ***link-state routing algorithm*** in which each router contains the information of every domain, and based on this information, it determines the shortest path.
- The goal of routing is to learn routes.
- The OSPF achieves by learning about every router and subnet within the entire network.
- Every router contains the same information about the network.
- The way the router learns this information by sending LSA (Link State Advertisements).
- These LSAs contain information about every router, subnet, and other networking information.
- Once the LSAs have been flooded, the OSPF stores the information in a link-state database known as LSDB.
- The main goal is to have the same information about every router in LSDBs.

OSPF Areas



- OSPF divides the autonomous systems into areas where the area is a **collection of networks, hosts, and routers**.
- Like internet service providers divide the internet into a different autonomous system for easy management, OSPF further divides the autonomous systems into Areas.
- Routers that exist inside the area, flood the area with routing information
- In Area, the special router also exists.
- The special routers are those that are present at the border of an area, and these special routers are known as **Area Border Routers**. This router summarizes the information about an area and shares the information with other areas.

- All the areas inside an autonomous system are connected to the **backbone routers**, and these backbone routers are part of a primary area. The role of a primary area is to provide communication between different areas.

How does OSPF work?

Step 1: The first step is to become OSPF neighbors. The two connecting routers running OSPF on the same link creates a neighbor relationship.

Step 2: The second step is to exchange database information. After becoming the neighbors, the two routers exchange the LSDB information with each other.

Step 3: The third step is to choose the best route. Once the LSDB information has been exchanged with each other, the router chooses the best route to be added to a routing table based on the calculation of SPF.

How a router forms a neighbor relationship?

The first thing that happened before the relationship is formed is that each router chooses the [router ID](#).

Router ID (RID): The router ID is a number that uniquely identifies each router on a network. The router ID is in the format of the IPv4 address. There are few ways to set the router ID, the first way is to set the router ID manually and the other way is to let the router decide itself.

- Two routers connected to each other through point to point or multiple routers are connected can communicate with each other through an OSPF protocol.
- The two routers are adjacent only when both the routers send the HELLO packet to each other.
- When both the routers receive the acknowledgment of the HELLO packet, then they come in a two-way state.
- As OSPF is a link state routing protocol, so it allows creating the neighbor relationship between the routers.
- The two routers can be neighbors only when they belong to the same subnet, share the same area id, subnet mask, timers, and authentication.
- The OSPF relationship is a relationship formed between the routers so that they can know each other.
- The two routers can be neighbors if at least one of them is designated router or backup designated router in a network, or connected through a point-to-point link.

Types of links in OSPF

A link is basically a connection, so the connection between two routers is known as a link.

There are four types of links in OSPF:

1. **Point-to-point link:** The point-to-point link directly connects the two routers without any host or router in between.
2. **Transient link:** When several routers are attached in a network, they are known as a transient link. The transient link has two different implementations:
 - Unrealistic topology: When all the routers are connected to each other, it is known as an unrealistic topology.
 - Realistic topology: When some designated router exists in a network then it is known as a realistic topology. Here designated router is a router to which all the routers are connected. All the packets sent by the routers will be passed through the designated router.
3. **Stub link:** It is a network that is connected to the single router. Data enters to the network through the single router and leaves the network through the same router.
4. **Virtual link:** If the link between the two routers is broken, the administration creates the virtual path between the routers, and that path could be a long one also.

OSPF Message Format

Version(8)	Type(8)	Message (16)
Source IP address		
Area Identification		
Chcek sum		Auth.Type
Authentication (32)		

- **Version:** It is an 8-bit field that specifies the OSPF protocol version.
- **Type:** It is an 8-bit field. It specifies the type of the OSPF packet.
- **Message:** It is a 16-bit field that defines the total length of the message, including the header. Therefore, the total length is equal to the sum of the length of the message and header.
- **Source IP address:** It defines the address from which the packets are sent. It is a sending routing IP address.
- **Area identification:** It defines the area within which the routing takes place.
- **Checksum:** It is used for error correction and error detection.

- **Authentication type:** There are two types of authentication, i.e., 0 and 1. Here, 0 means for none that specifies no authentication is available and 1 means for pwd that specifies the password-based authentication.
- **Authentication:** It is a 32-bit field that contains the actual value of the authentication data.

OSPF Packets in Open Shortest Path First (OSPF)

1. Hello Packet

- Used to **find neighbors** and check if they are alive.
- Helps to **start a connection** between routers.

“Are you there?” message.

2. Database Description (DBD)

- Sent after connection is formed.
- Contains a **summary of network topology**.
- Helps the new router **understand the network**.

“This is what my network looks like.”

3. Link State Request (LSR)

- Sent when a router needs **more details** about some routes.
- One router asks another for specific information.

“Give me more details.”

4. Link State Update (LSU)

- Used to **send detailed link information**.
- Router shares its **full routing data**.

“Here is the complete information.”

5. Link State Acknowledgment (LSAck)

- Confirms that updates are **received successfully**.
- Makes communication **reliable**.

“I received your message.”

OSPF States (Step-by-Step)

1. Down State

- No HELLO packet received yet.
- OSPF has not started.

“No communication.”

2. Init State

- Router **receives HELLO** from another router.

“I hear you.”

3. 2-Way State

- Both routers **exchange HELLO packets**.
- Neighbor relationship is formed.

“We can talk now.”

4. Exstart State

- Routers decide:
 - **Master**
 - **Slave**
- Based on **Router ID**

“Who will control the process?”

5. Exchange State

- Routers exchange **database summaries (DBD packets)**.

“Sharing basic information.”

6. Loading State

- Routers request and receive **missing information** using:
 - LSR
 - LSU
 - LSA

“Getting complete details.”

7. Full State

- All information is exchanged.
- Routers are fully synchronized.

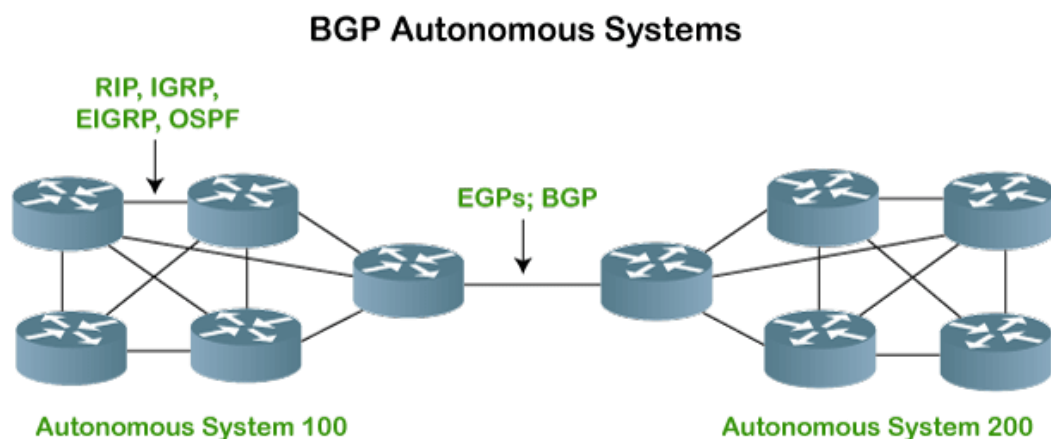
“Everything is updated.”

1. Hello → Find neighbor

2. 2-Way → Connection formed
3. Exstart → Choose master/slave
4. Exchange → Share summaries
5. Loading → Get full details
6. Full → Ready to route

3. BORDER GATEWAY PROTOCOL

- It is an **inter-domain routing protocol**, and it uses the **path-vector routing**.
- It is a gateway protocol that is used to exchange routing information among the autonomous system on the internet.
- As we know that Border Gateway Protocol works on different autonomous systems, so we should know the history of BGP, types of autonomous systems, etc.
- There are many versions of BGP, such as:
 - BGP version 1: This version was released in 1989 and is defined in RFC 1105.
 - BGP version 2: It was defined in RFC 1163.
 - BGP version 3: It was defined in RFC 1267.
 - BGP version 4: It is the current version of BGP defined in RFC 1771.



- An autonomous system is a collection of networks that comes under the single common administrative domain or we can say that it is a collection of routers under the single administrative domain.
- For example, an organization can contain multiple routers having different locations, but the single autonomous number system will recognize them.
- Within the same autonomous system or same organization, we generally use **IGP (Interior Gateway Protocol)** protocols like RIP, IGRP, EIGRP, and OSPF.
- Suppose we want to communicate between two autonomous systems. In that case, we use **EGP (Exterior Gateway Protocols)**.
- The protocol that is running on the internet or used to communicate between two different autonomous number systems is known as **BGP (Border Gateway Protocol)**.
- The BGP is the only protocol that is running on the internet backbone or used to exchange the routes between two different autonomous number systems. Internet service providers use the BGP protocol to control all the routing information

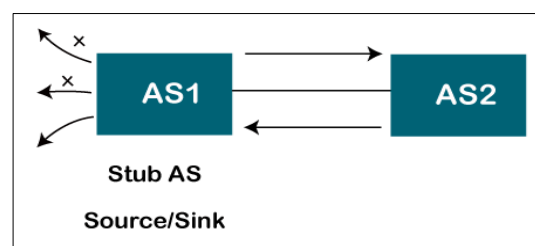
BGP Features:

Open standard: It is a standard protocol which can run on any window device.	Incremental and trigger updates: Like IGP, BGP also supports incremental and trigger updates.
Exterior Gateway Protocol: It is an exterior gateway protocol that is used to exchange the routing information between two or more autonomous system numbers.	Path vector protocol: The BGP is a path vector protocol. Here, path vector is a method of sending the routes along with routing information.
Inter AS-domain routing: It is specially designed for inter-domain routing, where inter AS-domain routing means exchanging the routing information between two or more autonomous number system.	Configure neighborhood relationship: It sends updates to configure the neighborhood relationship manually.
Supports internet: It is the only protocol that operates on the internet backbone.	Application layer protocol: It is an application layer protocol and uses TCP protocol for reliability.
Classless: It is a classless protocol.	Metric: It has lots of attributes like weight attribute, origin, etc. BGP supports a very rich number of attributes that can affect the path manipulation process.
Administrative distance: If the information is coming from the external autonomous system, then it uses 20 administrative distance. If the information is coming from the same autonomous system, then it uses 200 administrative distance.	

Types of Autonomous systems:

1. Stub autonomous system

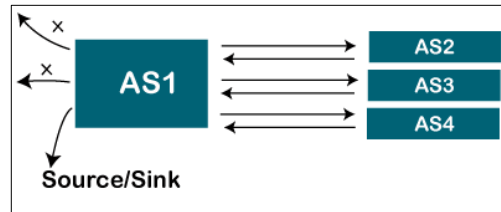
- It is a system that contains only one connection from one autonomous system another autonomous system.
- The data traffic cannot be passed through the stub autonomous system.
- The Stub AS can be either a source or a sink.
- If we have one autonomous system, i.e., AS1, then it will have a single connection to another autonomous system, AS2. The AS1 can act either as a source or a sink. If it acts as a source, then the data moves from AS1 to AS2. If AS1 acts as a sink, means that the data gets consumed in AS1 which is coming from AS2, but the data will not move forward from AS1.



to

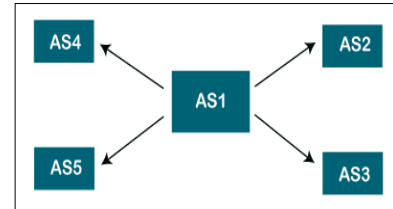
2. Multi-homed autonomous system

- It is an autonomous system that can have more than one connection to another autonomous system, but it can still be either a source or a sink for data traffic.
- There is no transient data traffic flow, which means that the data can be passed from one autonomous system.

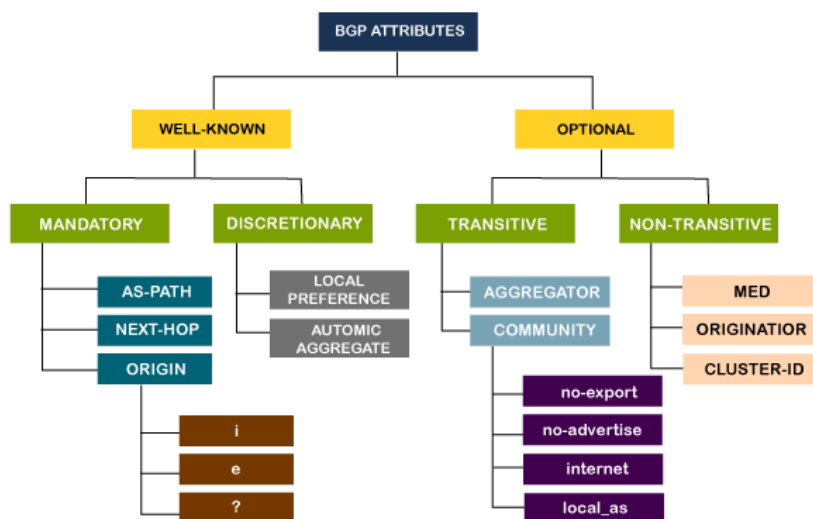


3. Transient Autonomous System

The transient autonomous system is a multihomed autonomous system, but it also provides transient traffic flow.



Path attributes:



- The BGP chooses the best route based on the attributes of the path.
- The path attributes provide the path information.
- The attributes that show or store the path information are known as path attributes.
- This list of attributes helps the receiving router to make a better decision while applying any policy.
- The path attribute is broadly classified into two categories:

1. Well-known attribute: It is an attribute that should be recognized by every BGP router.

The well-known attribute is further classified into two categories:

- **Well-known mandatory:** When BGP is going to advertise any network, but it also advertises extra information, and that information with path attributes information. The information includes AS path information, origin information, and next-hop information. Here, mandatory means that it has to be present in all the BGP routing updates.

- **Well-known discretionary:** It is recognized by all the BGP routers and passed on to other BGP routers, but it is not mandatory to be present in an update.

2. Optional attribute: It is an attribute that is not necessarily to be recognized by every BGP router. In short, we can say that it is not a mandatory attribute.

The optional attribute is further classified into two categories:

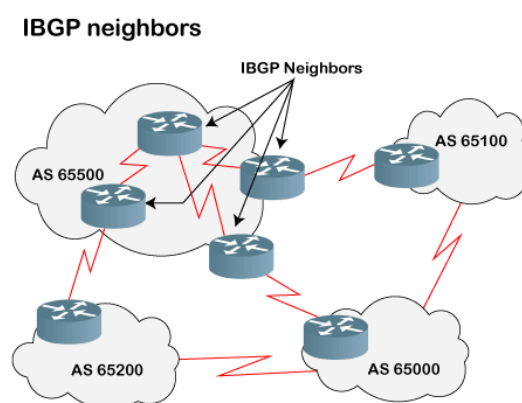
- **Optional transitive:** BGP may or may not recognize this attribute, but it is passed on to the other BGP neighbors. Here, transitive means that if the attribute is not recognized, then it is marked as a partial.
- **Optional non-transitive:** If the BGP cannot recognize the attribute, it ignores the update and does not advertise to another BGP router.

BGP Neighbors:

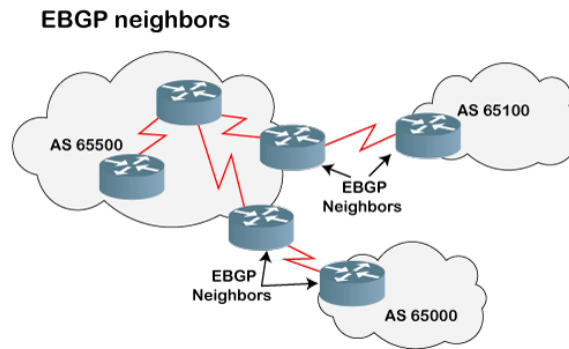
- BGP neighborship is similar to the OSPF neighborship, but there are few differences.
- BGP forms the neighboring relationship with the help of the [TCP](#) connection on port number 179 and then exchanges the BGP updates. They exchange the updates after forming the neighbor relationship.
- In BGP, the neighbor relationship is configured manually. BGP neighbors are also known as BGP peers or BGP speakers.

There are two types of neighbor relationship:

- IBGP (Internal BGP): If all the routers are neighbors of each other and belong to the same autonomous number system, the routers are referred to as an IBGP.



- EBGP (External BGP): If all the routers are neighbors of each other and they belong to the different autonomous number systems, then the routers are referred to as an EBGP.



BGP Tables:

- **Neighbor table:** It contains the neighbors who are configured by the administrator manually. The neighbor relationship has to be manually configured by using the neighbor command. For the verification, the following commands are used:

The above commands are very useful to verify whether the neighbor relationship is up or not.

- **BGP forwarding table:** It contains all the routes advertised in BGP and can be verified using the following command:

IP routing table: The IP routing table contains the best path routes required to reach the destination. The following command shows the best routing path:

BGP Sessions:

1. Internal BGP session

The internal BGP session is used to exchange information between the routers inside an autonomous system. In short, we can say that the routing information is exchanged between the routers of the same autonomous system.

2. External BGP session

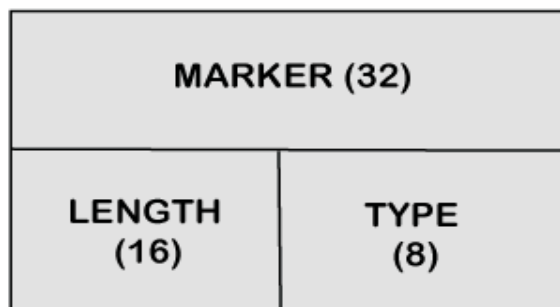
The external BGP session is a session in which nodes or routers of different autonomous systems communicate with each other.

Types of packets:

- **Open:** When the router wants to create a neighborhood relation with another router, it sends the Open packet.
- **Update:** The update packet can be used in either of the two cases:
 1. It can be used to withdraw the destination, which has been advertised previously.
 2. It can also be used to announce the route to the new destination.

- **Keep Alive:** The keep alive packet is exchanged regularly to tell other routers whether they are alive or not. For example, there are two routers, i.e., R1 and R2. The R1 sends the keep alive packet to R2 while R2 sends the keep alive packet to R1 so that R1 can get to know that R2 is alive, and R2 can get to know that R1 is alive.
- **Notification:** The notification packet is sent when the router detects the error condition or closes the connection.

BGP Packet Format:



1. **Marker:** It is a 32-bit field which is used for the authentication purpose.
2. **Length:** It is a 16-bit field that defines the total length of the message, including the header.
3. **Type:** It is an 8-bit field that defines the type of the packet.